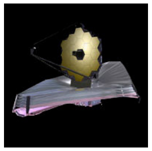

7729 - Unveiling Early Cosmic Enrichment: Direct Metallicities in z>6 Galaxies from Deep JWST Spectroscopy

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	gdn-dr5-final	NIRSpec MultiObject Spectroscopy	(3) gdn-final-dr5-precise
	2	GOODS-S	NIRSpec MultiObject Spectroscopy	(2) msa_targets_gds

ABSTRACT

What are the metal contents of galaxies, how quickly did they form, are strong-line diagnostics sufficiently robust to measure them? JWST has extended the frontier of galaxy metallicity studies, enabling measurements of strong diagnostics lines out to z~12. However, tracing the build up of metals across the first billion years remains extremely challenging and current constraints are sorely lacking. At z>6, simulations yield contrasting

predictions due to different feedback physics, while observationally only 17 objects observed with JWST have benefited from gold-standard, electron temperature-based metallicities. The vast majority of objects have instead relied on inferences using uncertain Oxygen and Balmer strong-line calibrations anchored to small numbers of 2<z<9 Te-based measurements. Comparisons between those calibrations indicate large discrepancies (up to 0.5-1 dex), casting serious doubt on the robustness of metallicities inferred thus far and indicating a need for stronger constraints.

This proposal aims to address these issues by determining direct metallicities for a statistical sample of confirmed z>6 galaxies in the GOODS fields, through deep NIRSpec G395M/F290LP spectroscopy of [O III]4963 and [O II]3728,3730. With a ~4x increase in detections compared to current samples, we will: (i) Establish the most robust R23 and O32 calibrations to date, and explore [Ne III]-based indices for applications at z>10. (ii) Utilize those calibrations to obtain robust metallicities in >500 sources at z=6-13 for mass-metallicity characterizations. (iii) Examine evolutionary trends and characterize the intrinsic scatter of galaxy metallicities over the first billion years.

OBSERVING DESCRIPTION

Pinpointing the onset and evolution of chemical enrichment over the first billion years represents a major goal of extragalactic astronomy and a key tracer of ISM evolution in primordial sources. Strong-line indices (e.g., R23 =[O III]4960,5008+[O II]3727,3729 / Hbeta, O32 =[O III]5008 / [O II]3727,3729) represent the most practical and realistic path towards obtaining a census of galaxy metallicities at z>6, given the strength and detection rates of the relevant lines in JWST spectra. However, even the latest JWST calibrations benefit from only a handful of gold standard, temperature-based ("direct") metallicities, owing to the dearth of auroral emission line measurements at z>6 (17 sources with [OIII]4364 to date). As such, current high-z calibrations suffer from significant scatter given small sample sizes (typically 15-20 sources) that extend over large redshift ranges (2<z<9) and trace varying ISM conditions.

In this proposal we target a large sample (>75) of confirmed z>6 galaxies in the GOODS-N and GOODS-S fields, where [O III]4364 (and [O II]3727,3729) detections are enabled for accurate Oxygen abundances using direct-method measurements. Each of the sources already benefit from strong-line measurements allowing us to make predictions on expected [O III]4364 line flux using realistic metallicity assumptions. We will obtain ~12.6 hrs of deep NIRSpec G395M/F290LP MSA spectroscopy for four independent pointings, two in each GOODS field amount to a total of 50.3 hrs science time, or 67.2 hrs including overhead. The chosen setup of an NRS readout pattern, 39 grps/int, 1 int/exp, and 27 exposures with a 3-shutter dither pattern allows us to achieve a peak S/N ratio >3sigma for integrated fluxes down to 1E-19 erg/s/cm2. The observations proposed here represents a unique, optimized, and sorely needed data set to achieve a census of galaxy metallicities within the first billion years.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	msa_targets_gds	RA: 03 32 38.8584 (53.1619100d) Dec: -27 47 7.40 (-27.78539d) Equinox: J2000		
Comments: *Description=[]*				
(3)	gdn-final-dr5-precise	RA: 12 36 46.7958 (189.1949825d) Dec: +62 14 54.23 (62.24840d) Equinox: J2000		
Comments: *Description=[]*				

Observation

Proposal 7729, Observation 1: gdn-dr5-final

Fri Apr 10 16:08:03 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	gdn-final-dr5-precise	RA: 12 36 46.7958 (189.1949825d) Dec: +62 14 54.23 (62.24840d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
2	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	Primary (144 sources)	Filler (801 sources)	jwst-nirspec-mr	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	10084	189.161869	62.247674	24.703	1	1008299	189.229448	62.249930	25.529
1	10117	189.235216	62.245169	24.541	1	1019607	189.159218	62.289641	24.260
1	10158	189.227536	62.260074	24.765	1	1029243	189.159935	62.260999	23.576
1	10171	189.219751	62.265081	24.909	1	1091033	189.269777	62.295885	25.105

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
2	10075	189.175448	62.219473	23.411	2	10100	189.146255	62.209070	23.111
2	10076	189.170408	62.209382	23.971	2	10106	189.107424	62.226893	23.803
2	10081	189.183564	62.249866	23.193	2	10162	189.188535	62.244937	24.906
2	10084	189.161869	62.247674	24.703	2	1005856	189.188893	62.242951	25.165

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395M/F290LP)	c1	3 Shutter Slitlet	189.20595641666668 Degrees 62.271743888888885 Degrees	275.1758835219529			3	21	27879.369
2	2 (G395M/F290LP)	c1	3 Shutter Slitlet	189.20595641666668 Degrees 62.271743888888885 Degrees	275.1758835219529			3	15	19913.835
3	1 (G395M/F290LP)	c2	3 Shutter Slitlet	189.13275174999998 Degrees 62.230395 Degrees	275.1112088908333			3	21	27879.369
4	2 (G395M/F290LP)	c2	3 Shutter Slitlet	189.13275174999998 Degrees 62.230395 Degrees	275.1112088908333			3	15	19913.835

Special Requirements

Group Visits within 53.0 Days
Visits Same PA
MSA Scheduled Aperture PA 275.1662 to 275.1662 Degrees (V3 136.59164 to 136.59164)

Observation

Proposal 7729, Observation 2: GOODS-S

Fri Apr 10 16:08:03 GMT 2026

Diagnostic Status: Error

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(GOODS-S (Obs 2)) Error (Form): This observation was created with an Aperture PA of 40.0000 but it has been assigned an Aperture PA of 48.7734
(Aperture PA) Error (Form): This observation was created with an Aperture PA of 40.0000 but it has been assigned an Aperture PA of 48.7734
(Visit 2:1) Error (Form): Reference stars are required but none were found for this visit
(Visit 2:2) Error (Form): Reference stars are required but none were found for this visit
(GOODS-S (Obs 2)) Warning (Form): Config c1 (#1) has 1 master background shutters affected by failed open or closed shutters.
(GOODS-S (Obs 2)) Warning (Form): Config c1 (#2) has 1 master background shutters affected by failed open or closed shutters.
(GOODS-S (Obs 2)) Warning (Form): Config c2 (#3) has 1 master background shutters affected by failed open or closed shutters.
(GOODS-S (Obs 2)) Warning (Form): Config c2 (#4) has 1 master background shutters affected by failed open or closed shutters.
(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	msa_targets_gds	RA: 03 32 38.8584 (53.1619100d) Dec: -27 47 7.40 (-27.78539d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	msa_targets_gds (207 sources)		jwst-nirspec-mr	1.5

Reference Stars

Proposal 7729 - Observation 2 - Unveiling Early Cosmic Enrichment: Direct Metallicities in z>6 Galaxies from Deep JWST Spectroscopy

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395M/F290LP)	c1	3 Shutter Slitlet	53.14469429166667 Degrees -27.78672027777776 Degrees	48.781442110930804			3	21	27879.369
2	2 (G395M/F290LP)	c1	3 Shutter Slitlet	53.14469429166667 Degrees -27.78672027777776 Degrees	48.781442110930804			3	15	19913.835
3	1 (G395M/F290LP)	c2	3 Shutter Slitlet	53.09333083333333 Degrees -27.783400555555545 Degrees	48.805392179384754			3	21	27879.369
4	2 (G395M/F290LP)	c2	3 Shutter Slitlet	53.09333083333333 Degrees -27.783400555555545 Degrees	48.805392179384754			3	15	19913.835

Special Requirements

Group Visits within 53.0 Days
Visits Same PA
MSA Scheduled Aperture PA 48.7734 to 48.7734 Degrees (V3 270.19885 to 270.19885)